

Compatibilidad $\mathcal{C}^\infty$ entre cartas

Definición 1 ($\mathcal{C}^\infty$ -compatibilidad). Sean (U_1, ψ_1) , (U_2, ψ_2) dos cartas de dimensión n en X , son $\mathcal{C}^\infty$ -compatibles $\iff$

- (i) $\psi_2 \circ \psi_1^{-1} \big|_{\psi_1(U_1 \cap U_2)} : \psi_1(U_1 \cap U_2) \longrightarrow \psi_2(U_1 \cap U_2)$ es $\mathcal{C}^\infty$.
- (ii) $\psi_1 \circ \psi_2^{-1} \big|_{\psi_2(U_1 \cap U_2)} : \psi_2(U_1 \cap U_2) \longrightarrow \psi_1(U_1 \cap U_2)$ es $\mathcal{C}^\infty$.

Ejercicio 1. Prueba que la $\mathcal{C}^\infty$ -compatibilidad es una relación de equivalencia.

Referenciado en

- prop-estructura-diferenciable-inducida-homeomorfismo
- prop-atlas-unicarta-imp-diferenciable
- atlas-diferenciable
- esp-proyectivo-real
- teo-existencia-unicidad-estructura-diferenciable
- lem-estructuras-diferenciables-iguales

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